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A digital twin approach for tunnel construction safety early warning and management

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ABSTRACT

Accidents tend to happen during tunnel construction due to the harsh environmental and geological conditions. Towards high safety risk during tunnel construction, a digital-twin based multi-information intelligent early warning and safety management platform is proposed in this study. At first, a communication line is built inside tunnel through multifunctional base stations. Multi-source information from tunnel construction could be collected and transmitted automatically. Then, dynamic tunnel construction information and surrounding rock conditions are virtual simulated. Meanwhile, four-level early warning and emergency response plan are built to provide judgement on the construction status. At last, application effects of the platform are validated in a deep buried tunnel project, Dongtianshan Tunnel in China, which has extreme weather, harmful gas, and bad geological conditions. A collapse accident of tunnel excavation face was successfully predicted by the platform. Early warning and appropriate emergency measures were taken in time, and the personnel safety was ensured.

1. Introduction

With the increase of tunnel numbers and capital investment, more deep-buried and extra-long tunnels were constructed (Ou et al., 2021). However, due to the bad construction environment (Zheng et al., 2012), complex geological conditions (Xu et al., 2020), and inadequate construction organization and management (Zhou and Ding, 2017), tunnel construction accidents happen frequently (Ding and Zhou, 2013; Zhou et al., 2015). For example, a water inrush accident occurred during tunnel construction with 14 persons dead, as shown in Fig. 1a. A gas explosion accident occurred during the railway tunnel construction, resulted in 12 deaths and 12 injured in the main tunnel, as shown in Fig. 1b.

To reduce the occurrence of construction accidents and avoid the casualties and property losses during tunnel construction, early warning is proposed and utilized in the safety management of underground engineering project (Lan et al., 2017; Salzano et al., 2009; Xing et al., 2019). Development of the technique of tunnel construction safety early warning mainly depends on two aspects: (i) monitoring means for the real-time safety information acquisition; (ii) information technology for dynamic multi-information transmission and fusion.

With the development of monitoring technique, the traditional manual monitoring and management methods are no longer suitable for the study of construction safety of underground engineering, instead, more technological means are involved (Xing et al., 2019; Jiang et al., 2022; Chen et al., 2020). Various monitoring means (Li et al., 2021; Wang et al., 2020), such as total stations (Zhou et al., 2020a), three-dimensional laser scanning (Jiang et al., 2021a), light detection and ranging (LiDAR) (Han et al., 2013a, 2013b), photogrammetry (Attard et al., 2018) are widely used for tunnel construction monitoring. However, as the tunnel construction information is diversified, variable and complicated, a single index is not enough for decision makers to use as the basis of judgment (Ding and Zhou, 2013; Liu et al., 2018; Fan and Zhou, 2021).

Meanwhile, the rapid development of information technology based on the internet allows for various possibilities in different aspects of

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(a) (b)

Fig. 1. Tunnel construction accident. (a) Search and rescue of water inrush accident; (b) Site of gas explosion accident.

tunnel construction. A term of Internet of Things (IoT) is widely applied in civil engineering due to its smarter and forward idea in the connection of physical and virtual part (Zhang et al., 2021a; Li et al., 2018). As an important part of realization of IoT, wireless sensor networks (WSNs) are employed in underground engineering to enhance safety and productivity and reduce operational costs. The improvement of ZigBee (Moridi et al., 2014), geographic information system (GIS) (Li and Li, 2014), general packet radio service (GPRS), and wireless technologies plays a certain role in the establishment of WSNs (Chen et al., 2020; Nizetic et al., 2020).

Based on this, through combination of information technologies and tunnel construction monitoring means, the intelligent early warning system is built and applied in various types of underground construction projects. Applications including excavations adjacent to the existing metro tunnels (Tian et al., 2021), prevention of risk of water and mud inrush (Wang et al., 2021a), tunnel cross passage construction under Yangtze River (Ding et al., 2013), settlement control (Fan and Zhou, 2021; Liu et al., 2020a), shield tunneling construction (Koch et al., 2017; Xie et al., 2018), earthquake (Fabozzi et al., 2018), urban tunnel construction for safety management (Ding et al., 2014), safety risk control for metro construction (Li et al., 2018), rock bursts in tunnel (Zhang et al., 2021b), underground mine (Jo and Khan, 2017), underground cavern construction (Ren et al., 2021), and preventing collisions inside tunnel (Baek and Choi, 2018).

The above applications promoted the progress of tunnel construction safety early warning system, but there are still some limitations. First, realization of multi-source information collection and transmission during tunnel construction is still a problem, especially under the harsh working environment and large buried depth (Zhou et al., 2015, 2020b; Salzano et al., 2009; Liu et al., 2020a; Ding et al., 2014; Zhang et al., 2021b; Pal and Hsieh, 2021). Second, most of studies are only focused on one aspect of the construction safety risks, and lack of combination with emergency plans and measures (Ding et al., 2013). Third, the entire tunnel construction is a dynamic, complex and long process, safety accidents happen randomly, but the current researches on dynamic virtual simulation of tunnel construction process are relatively deficient (Pal and Hsieh, 2021; Golovina et al., 2021). In addition, the term of "Digital-twin" (DT) has emerged in civil engineering (Jiang et al., 2021b). In DT, smart construction can be promoted and brought into the virtual world, and management can be conducted virtually at a low cost (Tuegel et al., 2011; Pan and Zhang, 2021; Boje et al., 2020; Schleich et al., 2017; Tao et al., 2019). However, the application of DT in the field of construction safety monitoring and management is still in its infancy (Jiang et al., 2021b; Boje et al., 2020), especially for tunnel construction safety early warning.

To fill this gap, the specific objectives of this research are as follows: (i) To propose a timely and effective feedback scheme for multi-dimensional dynamic construction information of tunnels under complex geological conditions; (ii) To solve the problem of risk monitoring and early warning lag in the harsh working environment of tunnel construction site; (iii) To explore the application effect of the safety management platform which combines the DT concept with the early warning system in tunnel construction.

Therefore, a DT-based multi-information intelligent early warning and safety management platform is proposed for tunnel construction to reduce the construction risks and avoid accident during tunnel construction in this study. The main contributions of this work are as follows: (i) Based on the combination of base stations with different functions, a multi-source dynamic information collection and transmission line for tunnel construction is constructed; (ii) The integration and bidirectional interoperation on a real-time basis between physical construction and digital construction in tunnel scenario is realized; (iii) A platform with four level early warning and emergency response plan, and remote interaction services for tunnel construction safety management in site is developed. The platform combines system, personnel, environment, and emergency plan to meet the dynamic design requirements of tunnel construction and ensure the safety of underground engineering construction. The DT-based platform was validated and verified in the project of Dongtianshan Tunnel in China. A collapse accident of tunnel excavation face was successfully predicted by the platform.

The rest of the paper is structured as follows. Related works on tunnel construction uncertainties, early warning and digital twin's potential use in the tunnel construction safety management are provided in Section 2. The architecture of a DT based multi-information intelligent early warning and safety management platform is described in Section 3. A case study in a practical tunnel construction project is performed in Section 4. Discussions are presented in Section 5. Conclusions and future works are outlined in Section 6.

2. Related works

2.1. Tunnel construction uncertainties

The expansion of underground space accelerated the development in tunnel engineering, particularly in China. Nevertheless, due to complex geographical environments, varied geological conditions, and incomplete safety management system during tunnel construction, which brings great uncertainty to the safety of tunnel construction (Sousa and Einstein, 2021; Ding et al., 2012; Guo and Zhang, 2021), frequent tunnel construction accidents can be caused (Zhou et al., 2022). To reduce and avoid tunnel construction safety accident, sufficient attention should be paid to the potential uncertainty in tunnel construction. The imprecision and randomness of tunnel construction is mainly featured in following aspects.

(i) In the aspect of human behavior and machinery, construction worker's physical quality and technical level can directly affect the construction safety (Ou et al., 2021; Zhou and Ding, 2017). Accident

records show that most accidents in civil engineering are caused by human faults. Human errors have repeated patterns and usually include lax management, poor technology, and improper handling of hazards. As for mechanical tunnel excavation, Liu et al. (2018) found that 30% of accidents are caused by human errors, while only 10% of the accidents could be completely attributed solely to natural causes, and 60% of accidents are resulted from the combination of human mistakes and natural disaster.

(ii) In the aspect of project management, it is reported that management and control errors are the causes of the accident (Zhou et al., 2015, 2020b; Ding et al., 2014), which including: inexperienced and improper management, inadequate emergency response measures, poor supervision of construction work, wrong construction method, errors in the collection and processing of monitoring data. Effective safety management can prevent the occurrence of accidents (Zhou et al., 2015). The International Tunneling Association (ITA) has published Guidelines for Tunneling Risk Management in 2004 (Ding et al., 2012). Meanwhile, a variety of evaluation methods have been proposed by many scholars (Ou et al., 2021; Zhou et al., 2020b), such as the fault tree method, artificial neural network methods, the method of fuzzy comprehensive evaluation, Bayesian network, and the multi-source information fusion, etc.

(iii) In the aspect of construction environment, the construction environment inside the tunnel is complex and harsh. The dim light and narrow space are not conducive to the workers and construction machinery activities (Zheng et al., 2012; Zhou and Ding, 2017; Wang et al., 2021a). The construction site ventilation is not smooth, the accumulation of harmful gases and dust is constantly threated the health of workers, and even cause adverse physical reactions. Noise from drilling holes and construction vehicles can lead to poor communication between workers in site, and sustained high intensity noise can also cause fatigue, resulting in low efficiency and high error rate. In addition, Sousa and Einstein (2021) proposed that unexpected ground conditions are often the main causes of accidents in tunnel construction, and classified the accident into rockfall, collapse, daylight collapse, flooding, spalling, excessive deformation, and collapse in specific locations.

Despite great efforts have made to avoid construction accidents, effective safety management remains a challenging task because of the significant uncertainties during the construction. For instance, Bjelland and Aven (2013) introduced the treatment of uncertainty in risk assessments in the Rogfast road tunnel project, and concluded that a much broader risk assessments of the uncertainties in the construction process is required. Zhu et al. (2022) statistically analyzed major 48 cases of tunnel construction accidents in China from 2010 to 2020, and pointed out that more than 60% of accidents are directly related to insufficient prospecting, poor monitoring, and lack of rescue planning. Therefore, it is very important to control multi-dynamic construction information comprehensively and timely, and formulate scientific construction early warning and response scheme for tunnel construction safety.

2.2. Early warning and its applications

Early warning system (EWS) is a warning system to mitigate the risks, which is composed of information communication systems and sensors, event detection and decision subsystems (Waidyanatha, 2010). In 1868, EWS was first used for earthquake disaster prevention in San Francisco (Nakamura et al., 2011). With the development of society, EWS has been gradually applied in the fields of military defense, economic, natural disasters and diseases, climate. Multiple sources of information are the foundation of building EWS. Lan et al. (2017) proposed that “multisource data are collected from multiple sources to meet one common surveillance purpose” in the summarization of development of early warning models. Yager (2016) claims out that “the early warning process is essentially a multi-source information fusion process, which conducts information collection, processing, screening, extraction and synthesis on the available data and then the judgment and decision could be made from the fused information”. Wang et al. (2021b) established a multi-factor comprehensive early warning identification model for real-time monitoring and early warning of mine water inrush.

Nowadays, numerous scholars have contributed in-depth studies on the EWS in the tunnelling engineering field. The traditional tunnel monitoring mainly relies on manual, and the prediction and early warning effect of tunnel accidents is relatively poor (Liu et al., 2020a). For the monitoring and early warning of the complex underground working area, on the one hand, intelligent monitoring system is established through sensors and wireless transmission. For instance, Wei Tian, et al (Tian et al., 2021). utilized a backpropagation neural network to predict the real-time deformation of the tunnel based on monitoring data from the adjacent construction site. Wang et al. (2021a) studied an improved method for the accurate measurement of sensors and wireless transmission bands suitable for complex tunnel environments, and proposed a real-time monitoring system for water inrush in tunnels. Liu et al. (2020a) designed an innovative intelligent tunnel monitoring system based on micro-service architecture and applied in the Anping tunnel. Zhang et al. (2021b) reviewed early warning methods about rockbursts in tunnel engineering based on microseismic monitoring.

On the other hand, real-time interaction and safety management are realized by wireless communication technology. Ding et al. (2013) presented a real-time safety early warning system to prevent accidents and improve safety management in underground construction, based on the IoT technology. Koch et al. (2017) presented a tunnel information modelling framework to support management. Xie et al. (2018) proposed a platform with timely auto-sending of monitoring data for settlement control during shield tunnelling construction. Jo and Khan (2017) introduced a real-time monitoring, event-reporting and early-warning platform based on the IoT for underground coal mines.

However, most of those studies are only focused on the recognition of potential specific risk, or identification of one specific aspect of the tunnel construction potential safety hazards, lack of and combination with emergency plans and measures (Ding et al., 2013). Considering entire tunnel construction project as a complex and dynamic process, information of tunnel construction is hard to completely controlled due to the signal shielding inside tunnel especially for large buried depth tunnel. How to construct information network and realize the transmission and integration of multi-information in tunnel construction is still a problem that needs to be solved at present.

2.3. Application of DT in tunnel construction

Nowadays, many scholars have defined and discussed the concept of DT (Tuegel et al., 2011; Pan and Zhang, 2021; Boje et al., 2020; Schleich et al., 2017; Hou et al., 2020). Akanmu et al. (2014) pointed out that the nature of DT is to connect a physical artefact to its digital counterpart. Tao et al. (2019) proposed the idea that “DT must consist of five parts: physical part, virtual part, connections, data, and services”. Specifically, the physical part is the foundation of the virtual part, and mirrored by the virtual part; data and information could be transferred by the connection part; simulation, decision making, monitoring and control of the physical object are provided by the services part of DT; data drives the services part to enhance the entire system. Jiang et al. (2021b) summarized the DT definition and its constituents in civil engineering.

At present, with the development of IoT, the concept of DT is applied in construction, operation and maintenance of the project. At the project construction stage, through application of DT in construction sites, construction information including process, machinery and materials, workers, quality and safety can be controlled and managed in real-time (Boje et al., 2020; Hou et al., 2020). For example, Pan and Zhang (2021) proposed an integrated digital twin framework for project management. Lei Hou, et al (Hou et al., 2020). summarized that although DT technology has been proven to be effective in improving construction worker safety, the construction industry still needs to take full advantage of

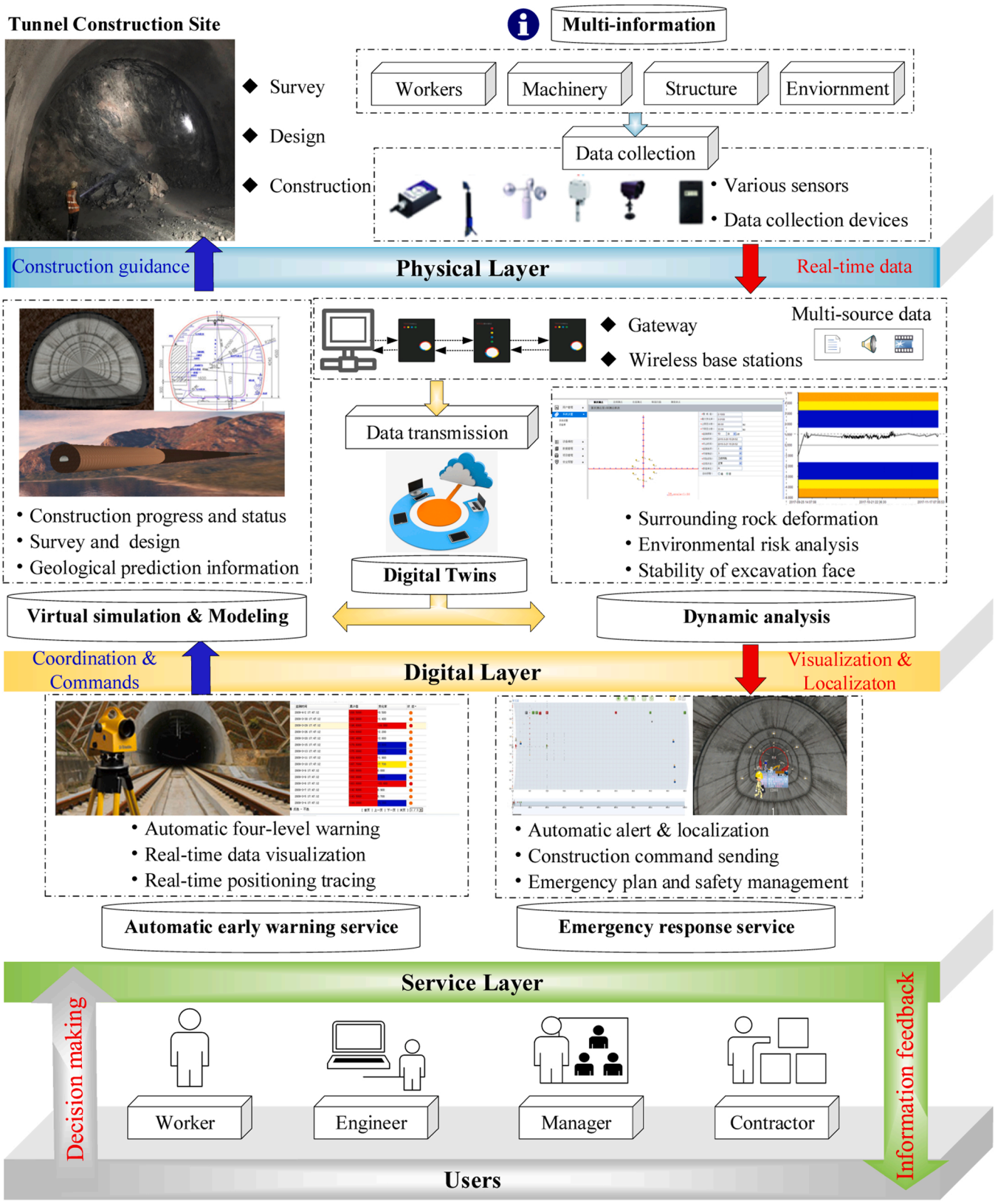


Fig. 2. DT-based platform framework.

these innovations in practice. Recently, application of DT approach in tunnel engineering is rare. The realization of DT approach for tunnels relies heavily on the data, namely the twin data which are extracted, from physical entities, virtual models and domain knowledge, and fused into each part simultaneously (Yu et al., 2021). Nevertheless, potential information of tunnel construction is difficult to obtain. Meanwhile, the data lacks deep integration and comprehensive interaction, and the real-time and accuracy of the tunnel construction management needs to

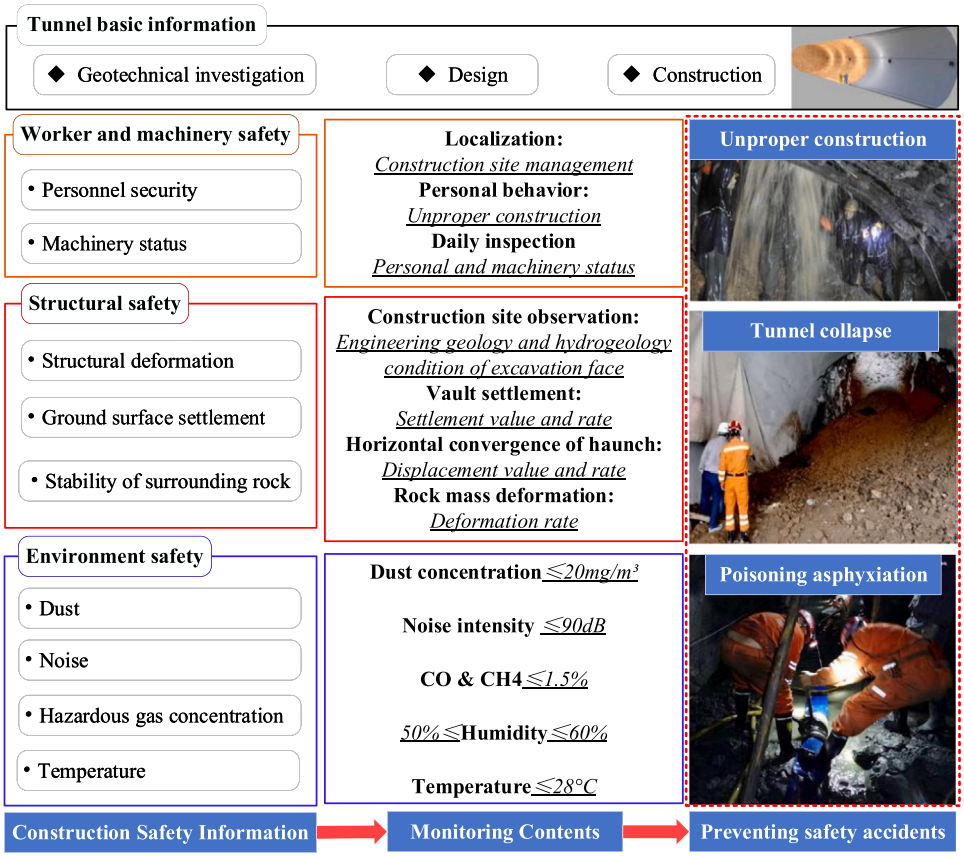


Fig. 3. Construction information and monitoring contents of the tunnel.

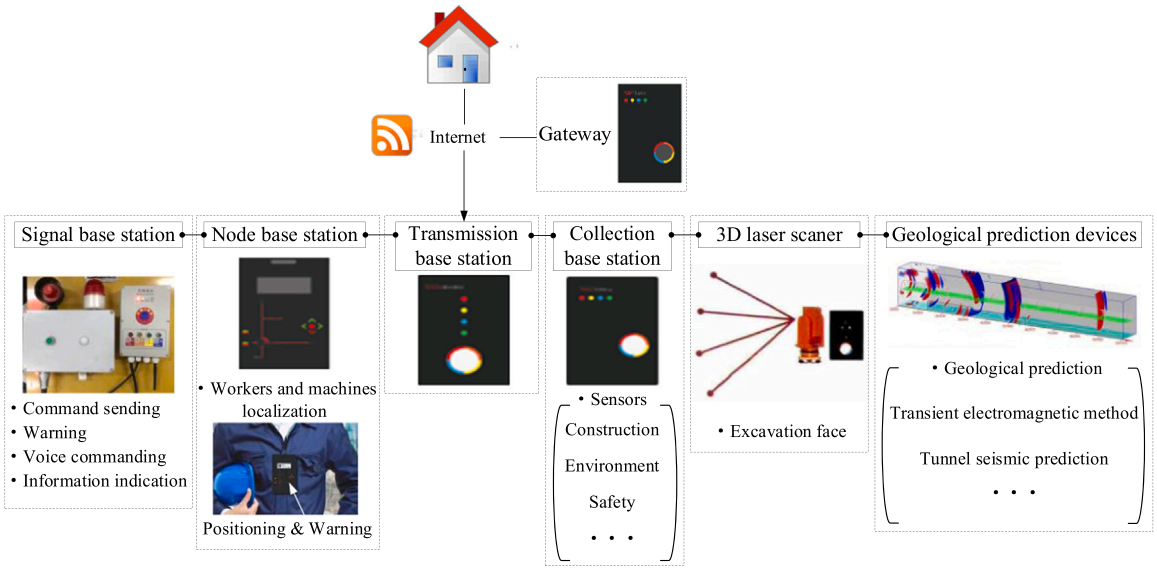


Fig. 4. Real-time data acquisition.

be improved.

Generally, with the application of DT, physical construction site can be brought into the virtual world, so that the simulation, calculation, analysis, optimization, and management can be conducted and operated remotely at a low cost (Qi and Tao, 2018). However, the application of DT in the field of construction safety early warning is almost "blank", and its application effect is desired to be explored. Therefore, based on the concept of DT, an intelligent early warning and safety management platform is established by integrating multi-source information from tunnel construction in this paper.

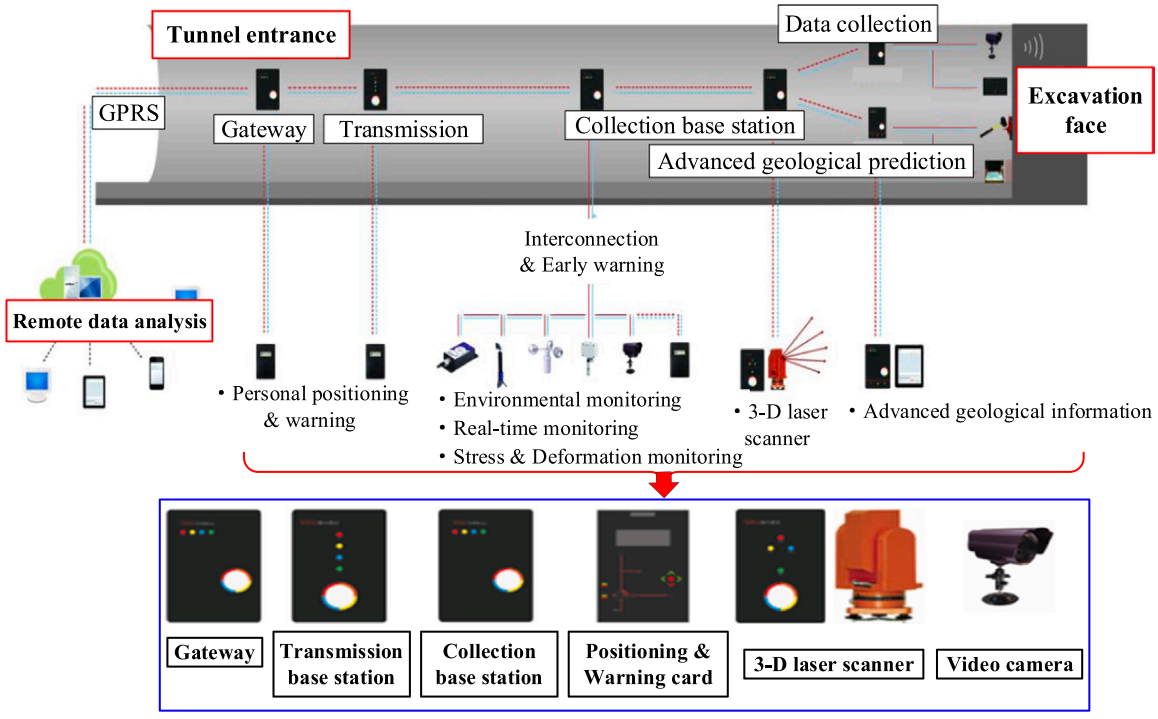


Fig. 5. Layout of hardware devices in the construction site.

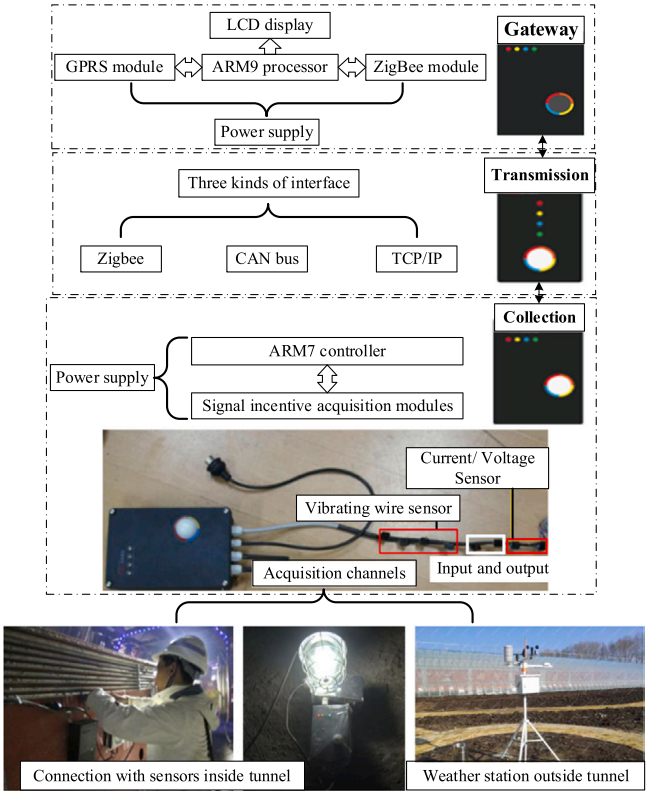


Fig. 6. Working principle and connections of equipment.

Table 1
Technical parameters.

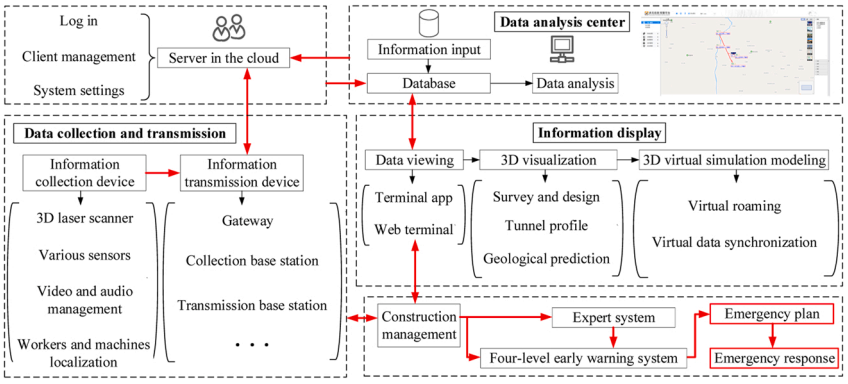
Index	Gateway	Base station	Positioning & warning card
FLASH	\	128KB	\
RAM	\	64KB	\
Frequency	2.4 GHz		
Maximum data rate	250 Kbps		
Sensitivity	-104 dBm		-87~– 92 dBm
Transmission distance	\	1200 m	10 m~100 m
Emission power consumption	\	< 290 mA	< 45 mA
Received power	< 50 mA		\
Sleep power	< 800 uA		
Working temperature	-20–70 °C		
Humidity	10–90%		
Power supply	220 V alternating current or battery		Battery
Seal capability	Waterproof		

3. Multi-information intelligent early warning and safety management platform

3.1. Design framework of the platform

To reduce construction safety risks and avoid emergency accidents, especially for adverse geological conditions and harsh environment, the platform in this paper was built based on the multi-information of tunnel construction, including information on tunnel survey, design and dynamic construction. Considering safety problems during tunnel construction, as we discussed in Section 2.1. DT technology is adopted at first time in the field of construction safety early warning and management in this paper.

According to the concept of DT, which consists of five parts: physical part, virtual part, connections, data, and services (Tao et al., 2019), Design framework of the platform as shown in Fig. 2. The working mechanism of the platform was as follows: at first, through the multi-function collection base station, construction monitoring data was



(a)



(b)

Fig. 7. Architecture of tunnel construction digital twins of the platform: (a) Process of DT; (b) Digital functions of the platform.

Table 2
Conversion of sensor test results.

Sensors	k	b	Formula (Technical Code for Monitoring Measurement of Railway Tunnel TB1-, 1012, 2007, 2007)
Humidity sensor	0.1	-18	$y = kx + b$
Dust concentration sensor	0.01	0	
Temperature sensor	0.13	0	
Illumination sensor	2.58	0	
Earth pressure cell	/	/	$y = kx^2 - kf_o^2 + N$
Reinforcement tensiometer	/	/	

automatically collected, the process triggered by specific events was recorded and analyzed, personal position was located, and tunnel excavation process was recorded. Then, those collected information was transmitted from base station to the data processing server by wireless. The collected data was processed, analyzed and feedbacked by the database. Multi-information of the construction site in time can be controlled by manager, and instructions and guidance for construction can be issued to the workers. Finally, a dynamic, interactive and intelligent way to manage tunnel construction site trend to be provided by the platform.

3.2. Physical layer

3.2.1. Multi-information of tunnel construction

Tunnel construction is a long and complicated process. Multi-information of tunnel construction should be controlled by manager in time to avoid safety accidents, as shown in Fig. 3.

3.2.2. Real-time data acquisition

To effectively obtain construction site information, real-time data acquisition is shown in Fig. 4. Several base stations with different functions are connected with the gateway to control and manage terminal equipment. Signal base station is responsible for commends sending, warning light control, voice commands and information cue. Node base station is used for positioning the workers and mechanical devices. Transmission and controlling base station carries out the uplink and downlink control of data and instructions. Collection base station is responsible for the collection and transmission of multi-information during tunnel construction. Various sensors, including safety sensors, environmental sensors, construction sensors, auxiliary sensors, are connected with collection base stations, so that real-time information about voice, video, image and data of the construction site could be gathered. Positioning and warning card is carried by workers at the construction site, which consists of induction, positioning and text message module. Personnel information could be identified and sent back to the service center as the person enters the base station detection range with the card, and distribution map and lists could be generated by the software, so that workers and machines position information could be directly seen by manager.

Layout of hardware platform in the construction site is shown in Fig. 5. The sensors and cameras are installed on the tunnel construction site, the laser instruments are arranged on the cross sections, the personnel cards are worn by the constructors, and all the devices are connected by wireless technology. The collected information is sent to the wireless base station with GPRS network through the wireless transmission link, and then the signal is transmitted to the monitoring center. A GPRS wireless transmission base station is erected at the tunnel portal, so that all the monitoring information can be sent to the monitoring center. Appropriate sensors are selected and installed according to the specific conditions of tunnel construction. All the sensors are uniformly controlled by the background server. Continuous power is supplied and sufficient battery capacity is provided to ensure 24 h normal operation of the system. Therefore, the management personnel can know about the tunnel safety and construction conditions anywhere through the Internet.

3.2.3. Communication line inside and outside tunnel

Real-time monitoring and detection are an important way to find out dangers during tunnel construction (Yao et al., 2021; Gou et al., 2021; Liu et al., 2020b; Ye et al., 2020, 2022; Greif et al., 2020). The working principle of the gateway and base stations are shown in the Fig. 6. Related technical parameters of gateway and multifunctional base stations are listed as shown in Table 1. Multi-information of the construction is obtained mainly by base stations that are arranged inside the tunnel. Meanwhile, the interfaces of Zigbee, CAN bus and TCP / IP protocols are provided by the transmission base station, so that various wired wireless networks can be constructed flexibly inside tunnel. Besides, various sensors can be connected with base station, including: ① digital signal, such as camera; ② currents and voltages signals, such as temperature, humidity, dust concentration; ③ frequency signal: all kinds of pressure cell, reinforcement gauge, etc.

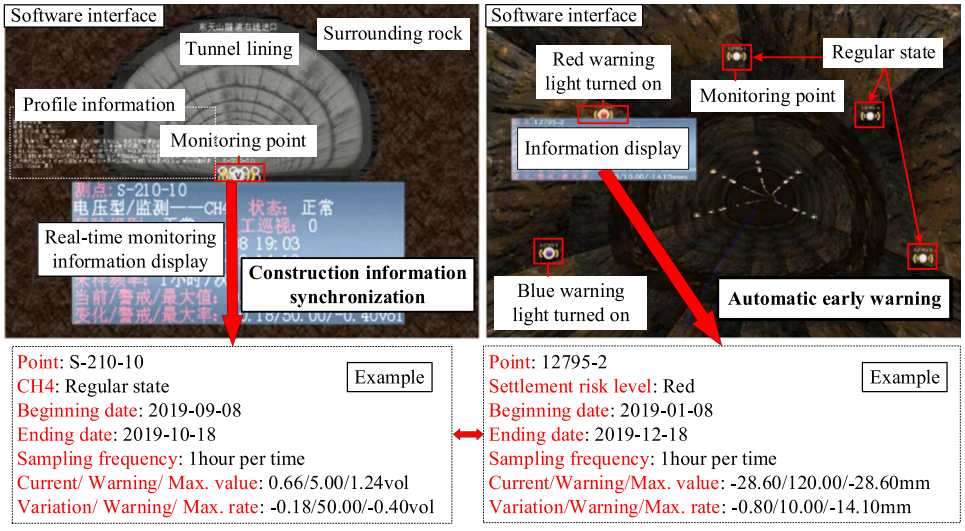


Fig. 8. Example of real-time construction data synchronization and automatic early warning.

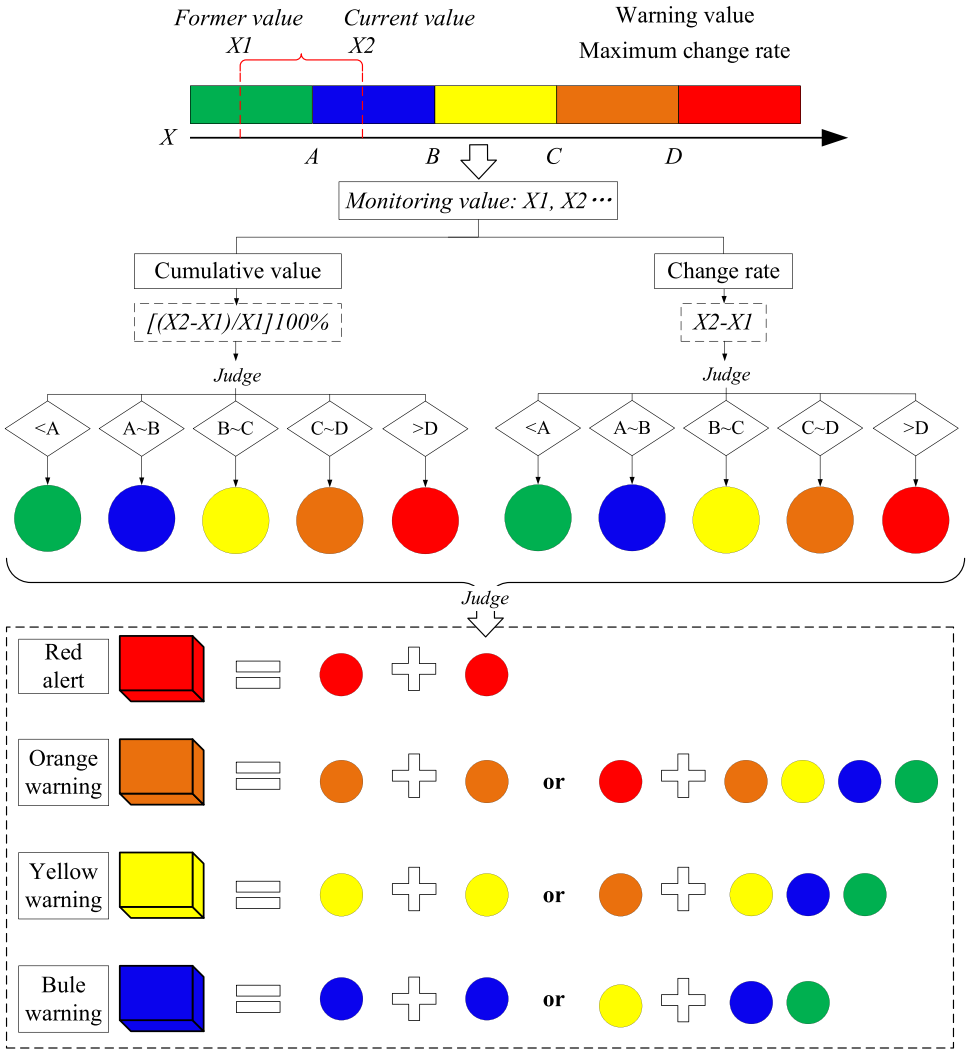


Fig. 9. Workflow of four-level warning.

3.3. Digital layer

3.3.1. Tunnel construction digital twins

Tunnel construction virtual model is constructed based on internet network equipment and real-time information collection. In order to make construction workers and managers operate conveniently and easily, Browser/Server (B/S) mode is selected (Ding et al., 2013). Process of DT model is constructed upon the real-time data on the construction site, as shown in Fig. 7a. Based on the characteristics of tunnel structure, tunnel engineering design scheme and structural parameters, tunnel simulation model, surrounding rock and environment model were established by Open Graphics Library (Open GL) technology, and tunnel investigation and design information, excavation, and lining construction process information were input and recorded. Digital layer of the platform is shown in Fig. 7b. Data visualization and implementation of alerts are important functions of the platform. Different with previous studies (Fan and Zhou, 2021; Tian et al., 2021; Ding et al., 2013; Liu et al., 2020a; Koch et al., 2017; Xie et al., 2018; Jo and Khan, 2017; Ren et al., 2021), the realization of intelligent early warning is a comprehensive process. By integrating multiple monitoring data, real-time updated advanced geological prediction information, construction progress and other information, the platform can not only realize the dynamic interaction between managers and workers, but also provide the basis for users to make comprehensive decisions.

In the scene prediction before the construction, as shown in Fig. 7b. At first, according to the feedback and update results of tunnel advanced geological prediction that collected in the platform, and combined with the current construction conditions of the tunnel, the layout position of monitoring sensors is determined. Then, the values of monitoring points are assigned in advance in light of the relevant specifications, structural numerical analysis results and expert experience. When the monitoring value exceeds the value assigned by the platform, early warning is started automatically, the corresponding emergency plan is triggered, and relevant manager and workers in site is informed.

Different sensors at the same measuring point are converted to the same physical quantity in the platform, correlation formulas are shown in Table 2. Thereinto, $y$is measurement results of multiple sensors, such as pressure, temperature, force, etc. x is the sensor collect data in field, such as value of current, voltage, frequency, etc. k is calibration coefficient, b is the constant that determined from sensor calibration curve. While, as for earth pressure cell and reinforcement tensiometer, f_0 is initial frequency, N is the corrected value. By presetting the threshold of measuring points in the platform, automatic warning and corresponding safety management functions can be realized.

3.3.2. Construction site roaming and real-time interaction

Construction real-time data and 3D model are synchronized based on monitoring results from tunnel construction site. Several monitoring points are established, such as environmental monitoring points of CH4, CO, temperature, humidity and illumination on the excavation face at

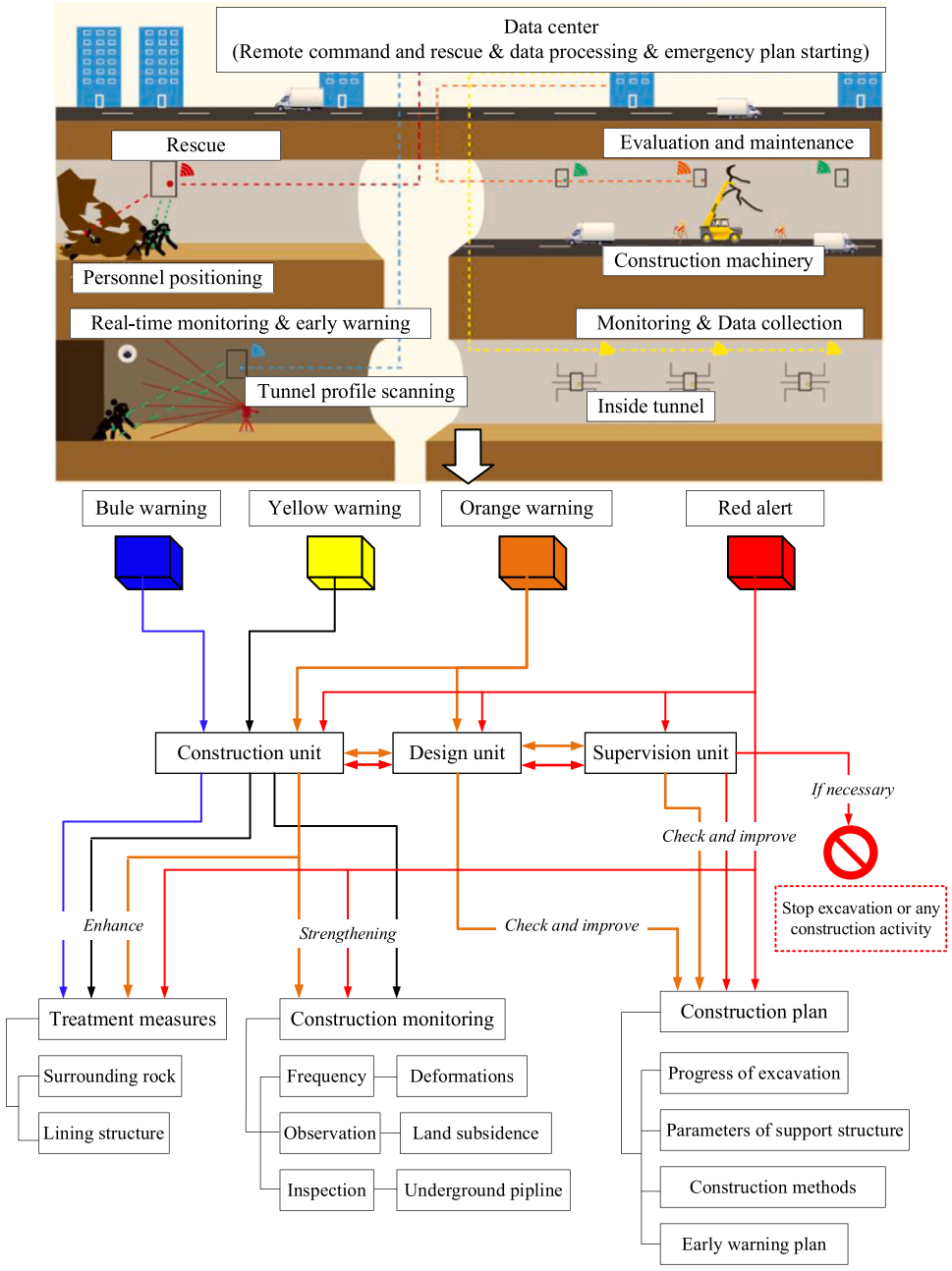


Fig. 10. Workflow of response plan.

the entrance, and meteorological monitoring points of wind speed, wind direction, illumination, temperature, humidity and atmospheric pressure near the entrance of the tunnel. The client could access the database on the server through the network, so as to obtain and display the real-time monitoring data on site. By setting the display parameters of the monitoring and measurement data, the managers could march and roam in the virtual tunnel.

Moreover, the data of the monitoring points could be viewed by opening the data display window at the mileage position of the monitoring points, one of the examples is shown in Fig. 8. Information about early warning level and content will be displayed in the platform when the monitoring data of a certain position on the construction site exceeds the warning value. Based on the four-level early warning mechanism, the corresponding response plan and related disposal measures could be released manually or automatically by computer. In addition, the indicators, whistle, personnel card prompt and SMS prompt of the base station at the construction site are associated with the four-level early warning system to provide the early warning and responses inside and outside the tunnel.

3.4. Service layer

3.4.1. Four-level automatic early warning service

To facilitate the users to judge the safety level of tunnel construction site, a four-level early warning is designed inside the software of the platform. Workflow of four-level early warning is shown in Fig. 9. Safety condition of the construction site could be reflected by the warning level. Based on the real-time monitoring value that transmitted from the construction site, the cumulative value and the change rate of monitoring point are both selected as control indexes. The warning level is determined by the color of those double control index. A single monitoring point may contain multiple aspects of tunnel construction

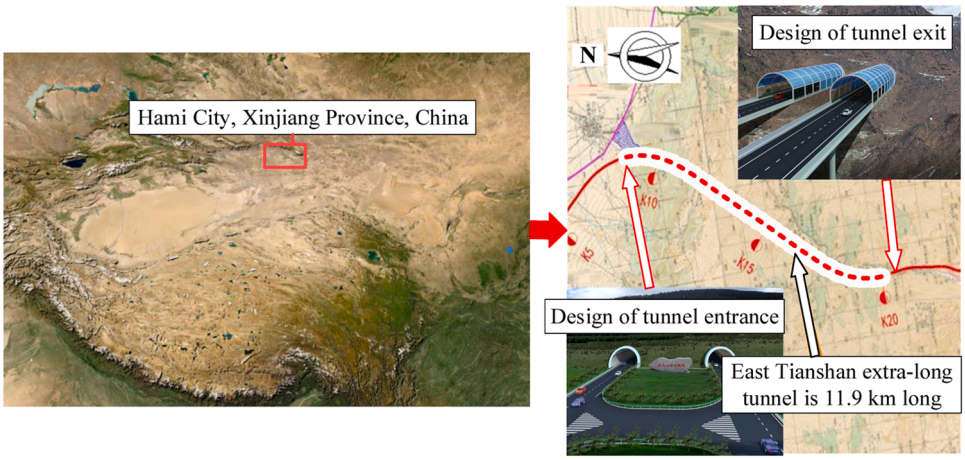


Fig. 11. Geographical location of the tunnel.

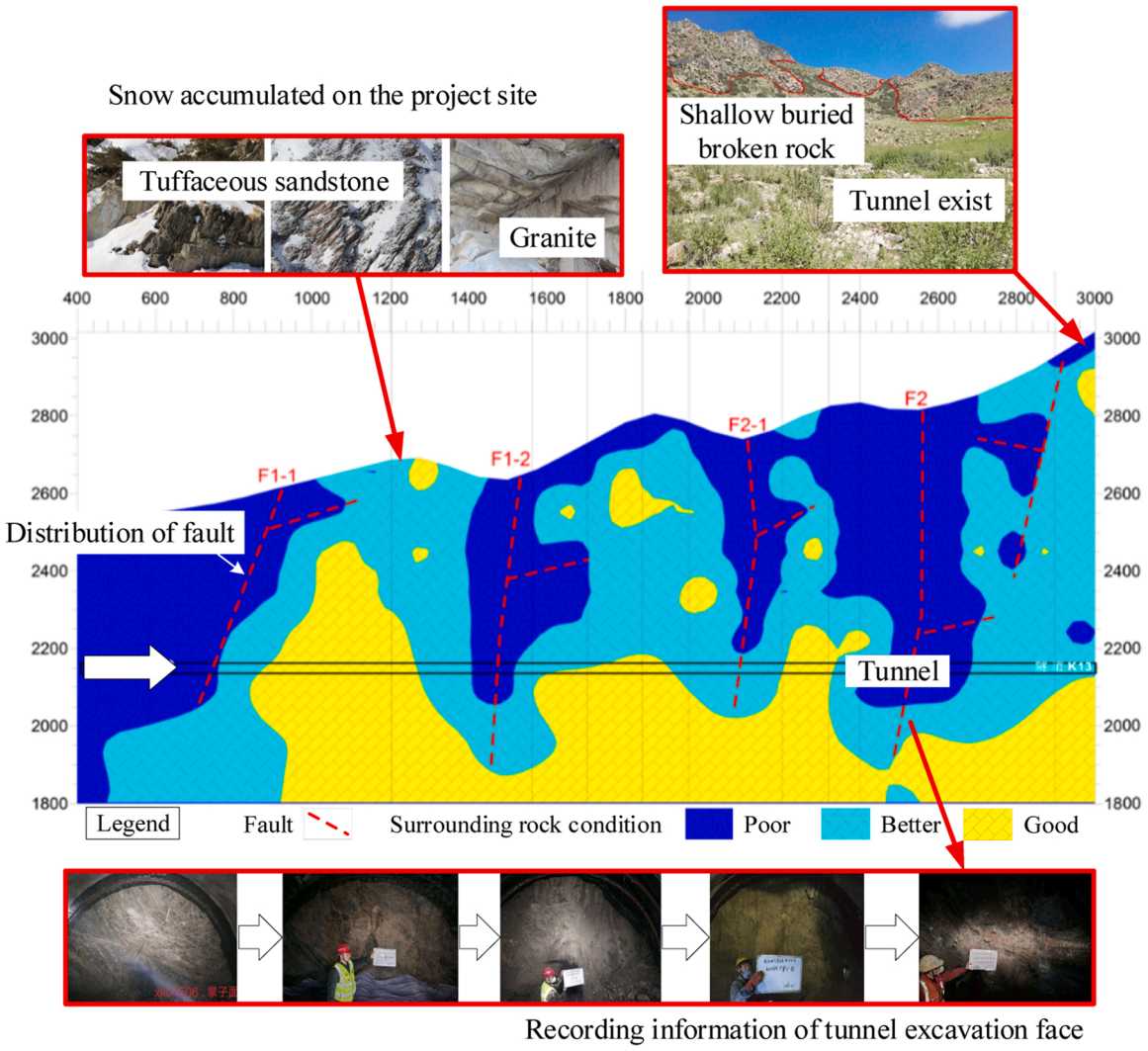


Fig. 12. Geological conditions of the tunnel site.

information, such as deformation of concrete structure, pressure from surrounding rock, and geological prediction. The value of the allowable maximum warning value and the maximum change rate can be input into the software by users according to the related risk management specification before the construction, then the value of four warning levels can be preset according to the project needs. Thus, the automatic early warning function of the monitoring items can be realized according to the preset standard value or threshold value of different measuring points and the risk level.

However, due to the complex tunnel construction environment and

Table 3
The geological conditions and recommendations.

Position	Surrounding rock types (Code for Design of Railway Tunnel TB3-, 1000, 2016, 2016)	Geological conditions	Recommend measures that should be strengthened
Tunnel entrance and exit	V	The exit of the tunnel is near the river; Mainly silty clay, pebble, silty clay; Underlying bedrock is tuffaceous sandstone; Poor stability.	Lining support; Drainage measures.
Tunnel body section	III~IV	Mainly tuffaceous siltstone; High geostress; Rock bust is easy to occur; Tunnel construction section is under the groundwater table.	Advance geological forecast before tunnel excavation.
Tunnel crossing fault zone	IV~V	Rock mass is mainly fractured moderately weathered tuffaceous sandstone; Mostly fragmented, and some are muddy; Joint fissures are extremely developed; Poor stability and easy to collapse.	Advance geological forecast; Lining support; Drainage measures; Safety management and emergency plan.

various construction data, it is difficult to avoid large differences or errors in monitoring results, resulting in delayed or even false positive early warning. False positives can result in impractical frequent stop of works and unwanted loss of money. To prevent this situation, after the red alert occurs, the platform will improve the monitoring frequency at the location of the monitoring point, or the corresponding manager can manually select and collect the data of the warning point through the platform, and judge the early warning status of the site according to the next test value. If the monitoring value back to normal, the platform will automatically return to the normal warning level, so as to avoid unnecessary personnel evacuation, construction stoppage and other economic losses.

3.4.2. Emergency response service

Response plan is established based on the four-level early warning to guide personnel evacuation, and a linked emergency response mechanism is formed as shown in Fig. 10. The software displays red, orange, yellow and blue, which correspond to four warning levels. The early warning level and content, the response measures, the supporting facilities (indicators of the base station, whistle and information signs at the portal), the prompts of personnel cards (vibration, indication, sound, SMS), personnel or unit responsibilities are integrated together, so as to build a linkage response inside and outside the tunnel.

4. Case study in Dongtianshan tunnel in China

4.1. Case study background

To verify the application effects of the platform proposed in previous section, the longest and highest tunnel project in Xinjiang, is selected as a case study to demonstrate the application effect. Dongtianshan tunnel is located in Hami, Xinjiang, China, its geographical location is shown in Fig. 11. The tunnel is 11.7 km long. The tunnel crosses the eastern part of Tianshan Mountain in Hami Region. It is designed with two inclined shafts, the total length of shaft 1# and its interconnection air duct is 2.5 km, and that of shaft 2# and its air duct is 1.5 km; and the distance between the two inclined shafts is 4.2 km.

The project site is located in a high-altitude area with complicated topographic, geological, hydrological and meteorological conditions, and high seismic intensity, where winters are cold, windy and snowy. Detailed geological and geotechnical conditions of the longitudinal

Table 4
Environmental sensors adopted in the project.

Sensors	Measurement range	Resolution ratio
CH4	0–500 ppm	1 ppm
CO	0–1000 ppm	1 ppm
Temperature	-50 ~ + 80 ℃	0.1 ℃
Humility	0 ~ 100%	0.1%

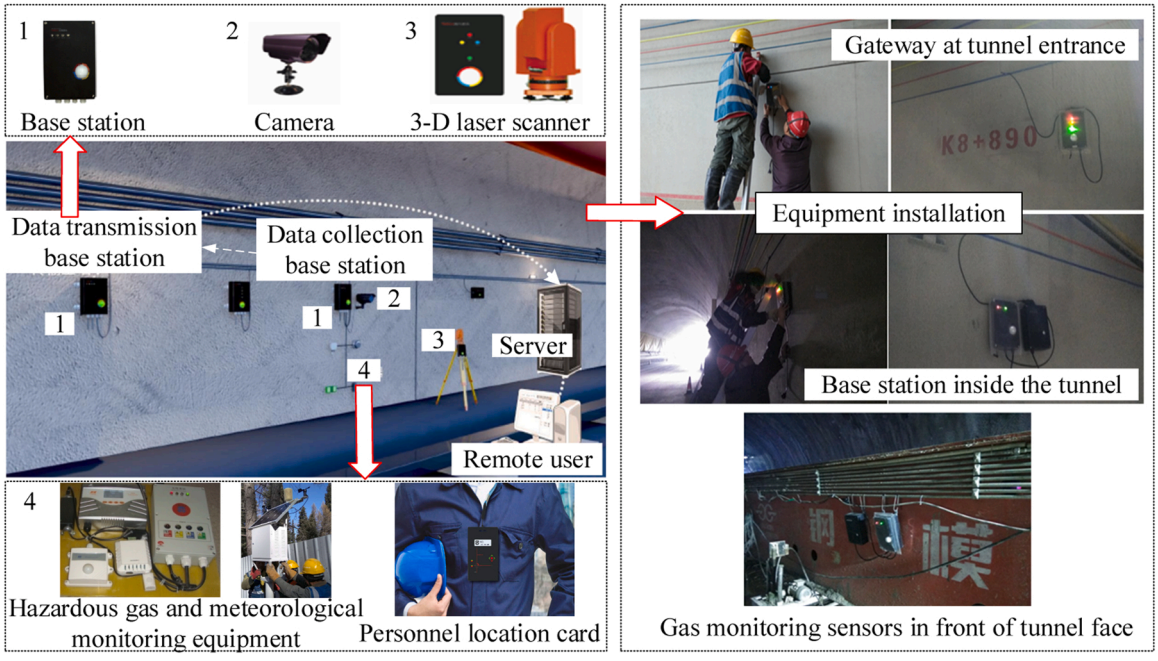


Fig. 13. System deployment and equipment installation.

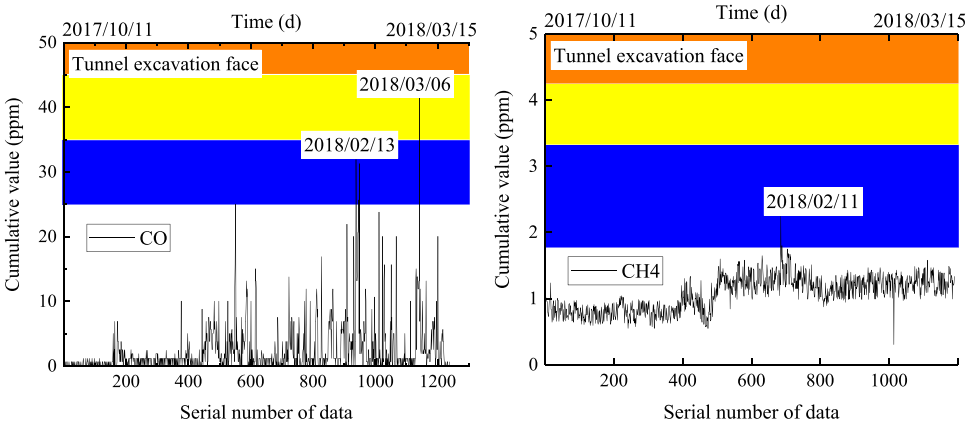


Fig. 14. Long-term monitoring data of harmful gas in working face.

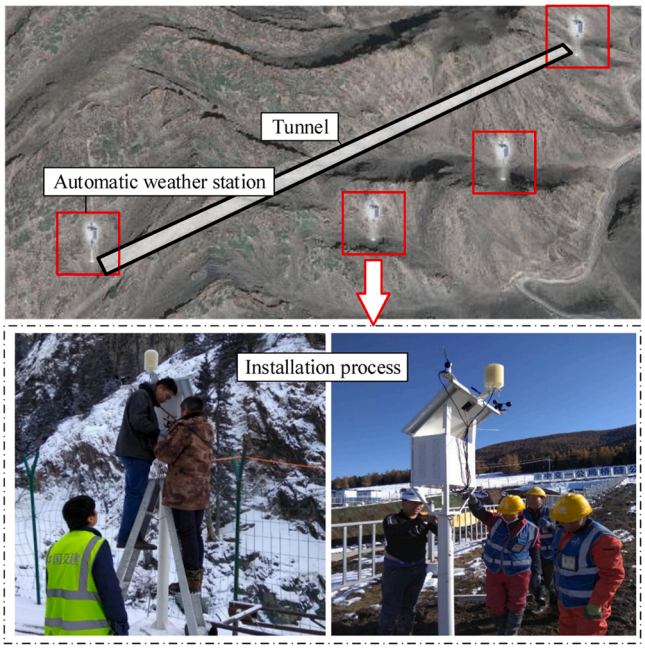


Fig. 15. Location of meteorological stations and site installation outside the tunnel.

cross-section of the tunnel are updated with the construction process. Face recording in front of tunnel excavation face is shown in Fig. 12. The tunnel passes through F1 active seismogenic faults in the Holocene, F2 inactive faults from north to south, with a length of 160 m, 320 m and 200 m respectively, a total of 680 m.

According to the geological investigation results, the overlying strata of the tunnel site are silty clay, pebble, gravel and block stone, while the underlying strata are tuff, tuffaceous sandstone, rhyolite, granite and pyroxite. The rock mass along the tunnel is relatively intact. However, due to the large buried depth of the tunnel and the influence of faults, the rock mass in some sections of the tunnel site is broken, and water inrush and collapse accidents are easy to occur. The geological and geotechnical conditions along the tunnel route are shown in Table 3. It is suggested that advance geological forecast and lining support should be timely strengthened, and corresponding emergency plan for construction safety should be properly adopted.

4.2. System deployment and equipment installation

System deployment and equipment installation are shown in Fig. 13. Specifically, the gateway base station was installed at the tunnel entrance where the network signal was relatively strong. In order to ensure the stability of wireless signal communication and the accuracy of personnel positioning, the data transmission base station was installed in the tunnel, of which the specific location was determined according to the strength of the wireless signal. Environmental sensors adopted in the project as shown in Table 4. A data acquisition base station and a harmful gas monitoring sensor were arranged on the excavation machinery near the tunnel face to monitor the concentration of carbon monoxide, gas and methane nearby.

4.3. Application effects

4.3.1. Harmful gas monitoring near the working face

Due to the long tunneling distance and the long fault crossing distance, it is very easy to produce harmful gas near the working face of the tunnel. Therefore, in the construction stage, the harmful gas in the tunnel environment was monitored in real time to judge the hazard of harmful gas to the constructors. The automatic monitoring data of carbon monoxide concentration and methane concentration on the left line of the tunnel, as shown in Fig. 14, indicated that the concentration of harmful gas increased during the construction stage, and sometimes even reached the blue and yellow warning value.

4.3.2. Automatic monitoring and collection of climate parameters under low temperature environment in Tianshan cold region

The Dongtianshan Tunnel is located in the Tianshan cold region with the altitude above 2000 m, the annual minimum temperature of − 32 ℃, the maximum freezing depth of 2.53 m, and the frost-free period of only 134 days. Therefore, the requirements for preventing and controlling frost damage and snowdrift, as well as the antifreeze and thermal insulation of tunnel, are high during the tunnel construction period, which lasted over four consecutive winters. In this application, four meteorological observation stations were established at the tunnel entrance, exit and the construction entrances of 1# and 2# inclined shafts, as shown in Fig. 15, so that the automatic monitoring of construction environment parameters such as wind speed, wind direction, temperature, humidity, illumination and atmospheric pressure are realized. The meteorological observation station was composed of a gateway base station, a data acquisition base station, a meteorological support, a solar cell pack and several sensors.

Fig. 16 shows the meteorological monitoring data at the tunnel entrance. The atmospheric pressure in the tunnel site during construction was relatively stable, with an average of about 80 kPa, which did not meet the early warning conditions, as shown in Fig. 16a. The wind speed was between 0 and 5 m/s in most of the time, and the maximum wind speed was about 12 m/s, slightly higher than the blue warning value, as shown in Fig. 16b. The temperature varied greatly throughout

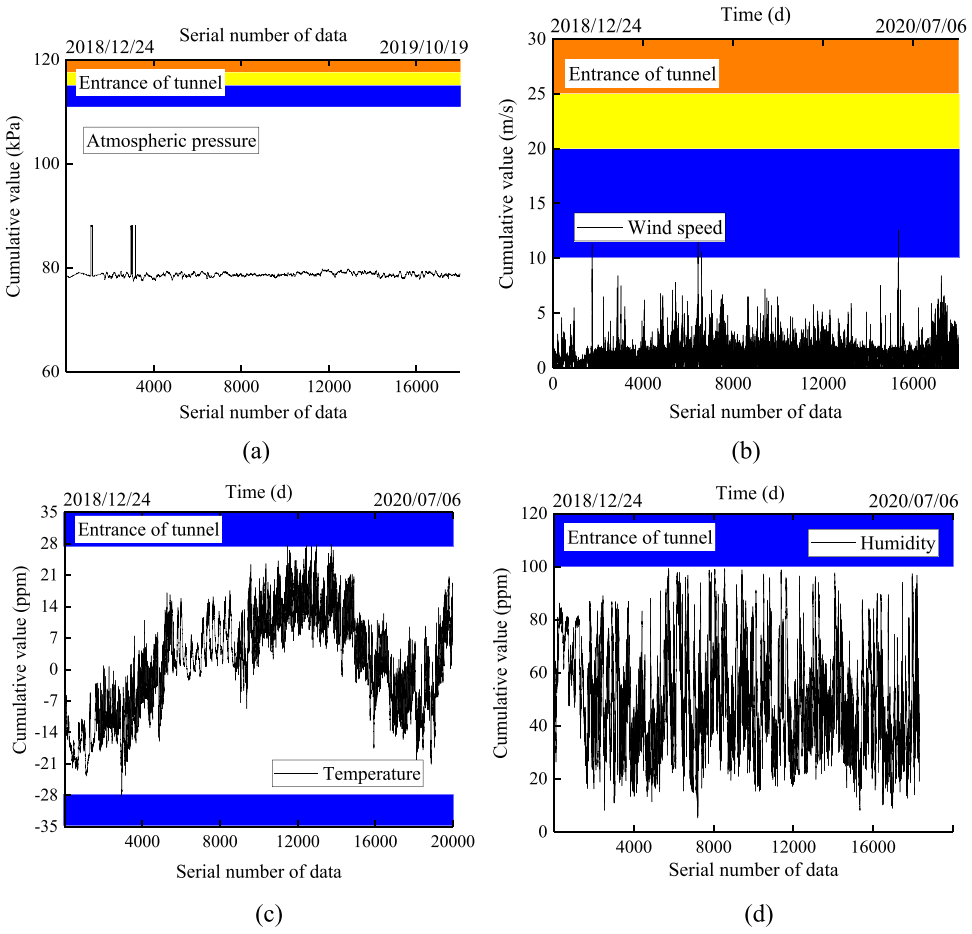


Fig. 16. Meteorological monitoring data: (a) Atmospheric pressure; (b) Wind speed; (c) Temperature; (d) Humidity.

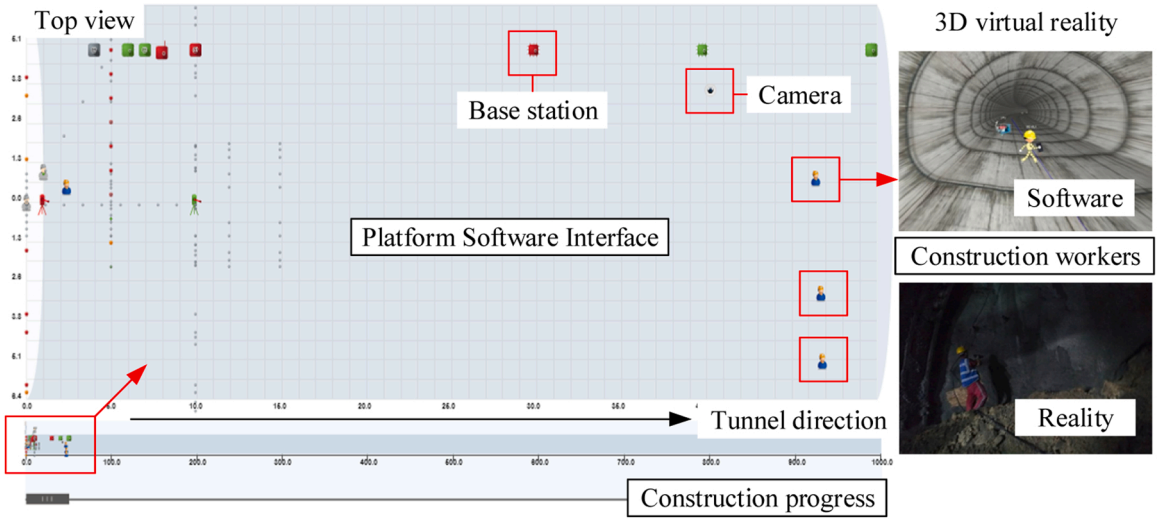


Fig. 17. Real-time positioning display of personnel and equipment.

the year, ranging from − 28–28 ℃, as shown in Fig. 16c. Additionally, the humidity also fluctuated greatly between 10~90%, as shown in Fig. 16d.

4.3.3. *Workers and machines positioning*

Due to the wide range of tunnel construction site, scattered personnel distribution and large communication blind area in the tunnel, it was difficult to locate the workers and operation track in real time. Data transmission base stations were equipped at the entrance and exit of the tunnel, crossings, working face, main entrance and exit on the ground, so that a person entered the detection range of the transmission base station with an identification card, his or her identity information could be read by the transmission base station and transmitted to the data center of the system. Then, a personnel distribution map, as shown in

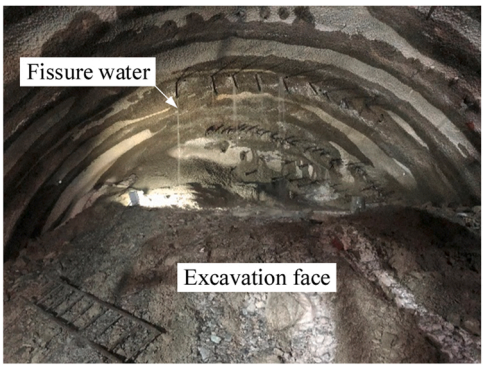


Fig. 18. Geological conditions in front of the tunnel face.

Fig. 17, was generated by processing the location information of the personnel card.

4.3.4. Safety risk early warning and management during construction

During the tunnel construction in the active fault F2, the platform launched a red early warning once. The tunnel face was then located at K0 + 668, as shown in Fig. 18. The middle and lower part of the tunnel face was mainly composed of the strongly-weathered tuffaceous sandstone, where the surrounding rock was very broken, the joint fissures were well developed, and the rock mass was mostly fracture. The fissure water ran out of the tunnel face in strands. The surrounding rock was not stable with weak bonding force between strata. The upper part of the tunnel face was mainly composed of the fully-weathered tuffaceous siltstone in mud shape, with water seepage on the surface and extremely unstable surrounding rock.

Due to high tunnel construction risks, the geological prediction and monitoring of surrounding rock deformation was strengthened in the construction site. The construction safety emergency and response plan were preset in the platform before the construction. Process of the response was shown in Fig. 19. At first, according to the information on advanced geological prediction, which includes the geological logging of the tunnel face, the tunnel seismic prediction and the exploration data of advanced horizontal drilling, in the 30 m section from K0 + 668 to K0 + 698, the surrounding rock was seriously broken, rich in water, and likely to collapse and have water inrush, so that the great attention should be paid. Then, based on real-time information of surrounding rock deformation, the platform gave out an early warning and prediction in time for the possible collapse during excavation. Meanwhile, an emergency plan was automatically launched by the platform to evacuate the workers on site in case the collapse and water inrush occurs. Finally, multiple messages receiving functions were provided through SMS, mobile client and web pages to build the connection between workers and managers, so as to realize the early detection and early treatment of safety abnormalities and alarms, prevent disasters, and reduce accidents.

The tunnel face collapse was recorded by the remote video camera, lasting about only 1–2 min, as shown in Fig. 20. However, due to the timely warning and the appropriate measures taken, no casualty was caused in the collapse. Subsequently, with reference to the geological conditions in construction at the fault, the contractor developed the relevant construction measures to prevent collapse, such as reserving the core soil on the tunnel face, lengthening the advanced small pipe appropriately and strengthening the grouting. In addition, the lining structure was reinforced to control the deformation strictly. Because of the real-time monitoring, feedback and early warning of the system during tunnel construction, no accident occurred during the whole process of tunnel construction.

According to the application effect of the platform in the collapse accident safety management during construction, the main contribution of the platform includes: (i) an orderly and timely communication line inside and outside the deep buried depth tunnel is established; (ii) the construction scene of digital twin tunnel is constructed, and early warning and emergency plan services are provided. The application of the platform gives a useful reference for the construction and safety management of tunnel engineering.

5. Discussions

Construction safety early warning is a necessary task for the entire tunnel construction management. Compared with traditional means and previous studies, the merits of the platform in this paper are summarized as follows: (i) Through the integration of different functional base stations and multi-source data, the effective transmission and collection of tunnel dynamic construction data are realized. (ii) Compared with previous studies in the construction safety early warning (Fan and Zhou, 2021; Tian et al., 2021; Ding et al., 2013; Liu et al., 2020a; Koch et al., 2017; Xie et al., 2018; Jo and Khan, 2017; Ren et al., 2021), the platform in this paper provides more services and functions, such as real-time image acquisition of tunnel construction face, long-term monitoring of entire construction process, early warning and safety management, etc. (iii) The construction of the platform is a beneficial attempt based on the DT concept. A 3D virtual tunnel model is established by integrating tunnel survey, design data and construction dynamic data. The virtual tunnel roaming function is developed to realize the virtual dynamic tunnel browsing and monitoring, as well as the real-time visualization display inside the virtual tunnel. (iv) The platform automatic warning classification method is developed, and the alarm threshold and grade can be preset by client. When the warning is issued, the location, level and other detailed information of the warning are dynamically displayed in the platform, and transmitted to the construction site to start the corresponding emergency response plan.

However, there are two limitations in the proposed platform. First, the details displayed in the construction process of virtual simulation part of the platform need to be strengthened. Detailed display and monitoring about worker's behavior and health (Zhou and Ding, 2017), and preventing collisions (Baek and Choi, 2018) in the construction site are useful to be added in the future. Second, how to analyze potential safety risk via various information and prevent accident before the construction is also important (Lan et al., 2017; Wang et al., 2021b). A large numbers of safety information were involved during tunnel construction, effective safety risk analysis based on the engineering geology and environment conditions of project before construction should be added to the platform.

6. Conclusions and future work

Towards high safety risk during tunnel construction, a DT based multi-information intelligent early warning and safety management platform is proposed in this study. A detailed framework of the DT containing physical, virtual, and connection, data, service parts are developed and introduced. The conclusions are as follows:

In the physical layer of the platform, a tunnel construction information collection and transmission line based on WSNs is established. Several base stations with different functions, including instruction sending, warning, positioning the workers and mechanical devices, information transmission, connected with the gateway to control and manage terminal equipment. Various sensors, including safety sensors, environmental sensors, construction sensors, auxiliary sensors, integrated and linked with the base stations of the platform. Dynamic construction information about voice, video, image and data of the construction site can be fluently transferred inside and outside the tunnel. Meanwhile, multi-source information of construction site can be automatically sent and received by the platform, including temperature, humidity, dust concentration near the excavation face, structural and surrounding rock deformation, updated advanced geological prediction. A real-time connection between construction manager and workers

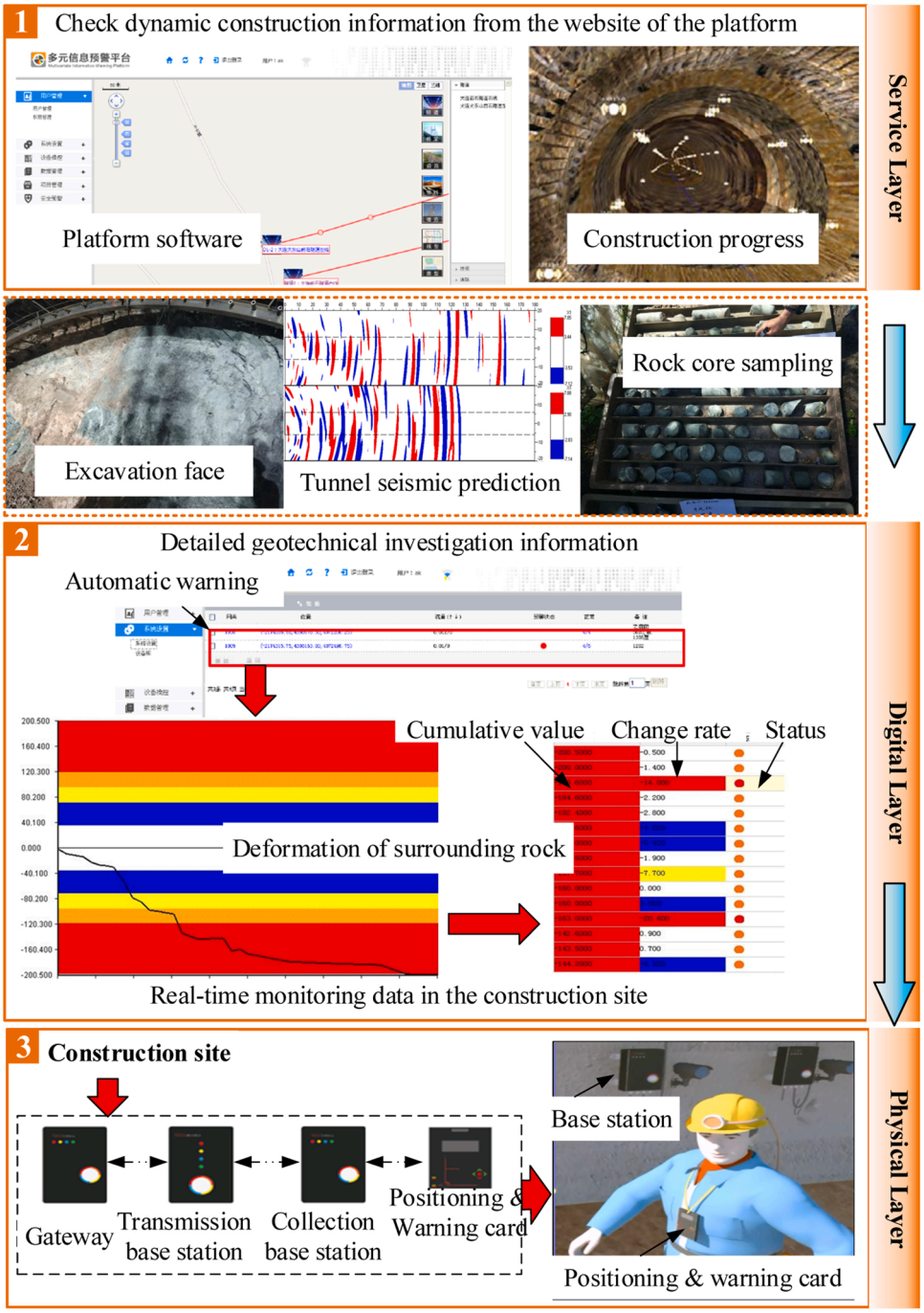


Fig. 19. Emergency plan and process of intelligent response.

during tunnel construction is established finally.

The digital layer of the platform is built upon the physical layer. In the digital layer of the platform, tunnel construction virtual model is constructed through the internet network equipment and real-time information collection device. Tunnel and its surrounding conditions are virtually simulated in 3D according to its geological investigation, design and dynamic construction information. In light of the monitoring results from tunnel construction site, construction real-time data and 3D model are synchronized. From this, actual physical situation of the construction site can be mirrored by tunnel construction virtual model. Moreover, the service layer of the platform is developed to help manager control real-time information of construction site. A four-level early warning and its corresponding emergency response plans are provided to judge the construction status and help construction manager make correct decision.

At last, the DT-based platform was validated and verified in the project of Dongtianshan Tunnel in China. To guide the safety management in the site and minimize potential uncertainties during tunnel construction, the platform is deployed and real-time synchronization of virtual tunnel and site construction data, 3D visualization display, and early warning and emergency management are realized. Specific functions include: (i) harmful gas monitoring; (ii) climate parameters automatic monitoring under low temperature environment; (iii) workers and machines positioning; (iv) safety risk early warning and management during construction. Finally, a collapse accident of tunnel excavation face was successfully predicted by the platform. Early warning and appropriate emergency measures were taken in time, and the personnel safety was ensured. Development of the platform provides a DT

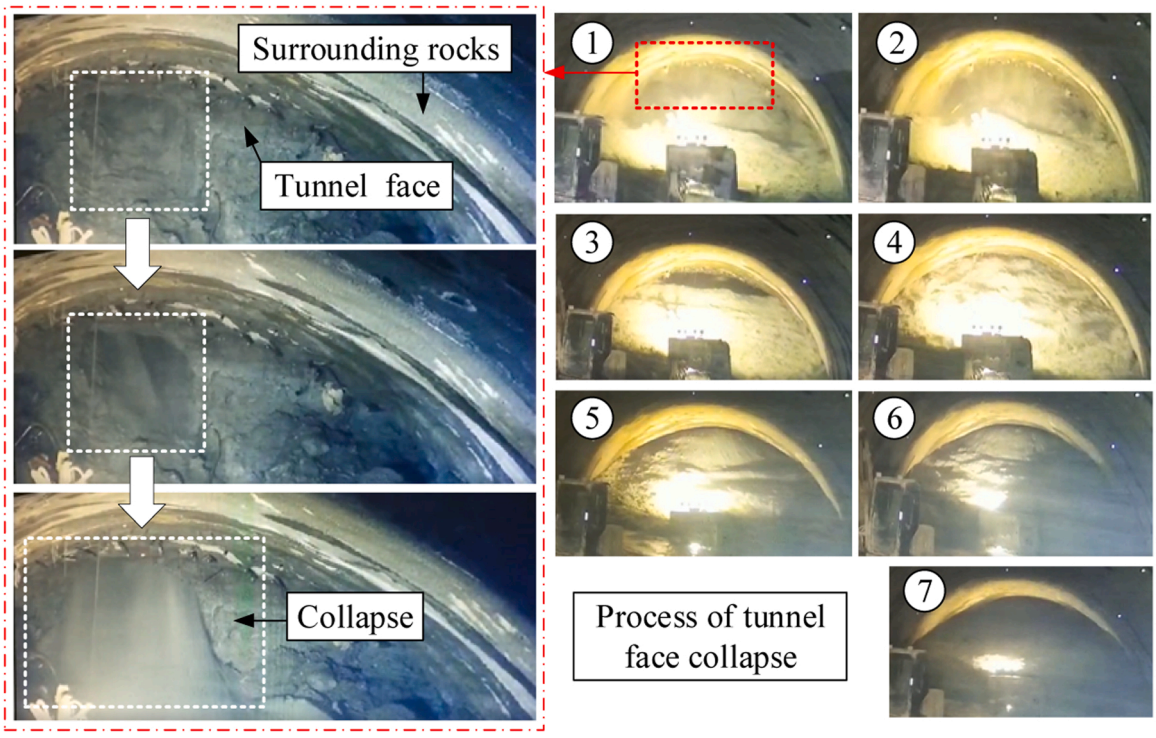


Fig. 20. Collapse of tunnel face recorded in the remote video camera of the system.

approach for tunnel construction safety early warning and management, especially for tunnel construction under the environment of complicate, low temperature, high safety risk, which has practical application value.

Nevertheless, the concept of DT has been put forward for many years, there are still a long way to go to realize digitalization and informatization of tunnel construction. The following possible work for this research includes: (i) it is necessary to further increase more kinds of monitoring data to better serve the safety management; (ii) a deeper risk analysis before construction should be supplied by the platform; (iii) database of tunnel construction information should be improved for further analytical analysis, since real-time monitoring data of entire construction process not only can be used for early warning and safety management during the construction, but also can be applied for predictive analysis in maintenance stage.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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